

Masapeta Kuruva Bhargava Sri Sai

Kurnool, Andhra Pradesh, India

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Professional Summary

Computer Science undergraduate specializing in backend engineering, distributed systems, and applied machine learning. Hands-on experience designing and shipping cloud-native microservices, high-performance Rust networking software, retrieval-augmented generation (RAG) pipelines, and real-time computer vision systems. Proven record of end-to-end ownership across 30+ open-source repositories, from system architecture and API design to CI/CD, containerization, and deployment. Seeking software engineering internship opportunities in backend, platform, or AI/ML engineering.

Education

Amrita Vishwa Vidyapeetham

Aug 2024 – May 2028

B.Tech in Computer Science & Engineering

Coimbatore, India

- Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Machine Learning

Technical Skills

Languages: Python, Rust, TypeScript, JavaScript, Java, C++, SQL, HTML, CSS

Backend & Systems: FastAPI, Node.js, Axum, Tokio, Hyper, REST APIs, Microservices Architecture, Reverse Proxy, API Gateway, Asynchronous Programming, Concurrency, WebSockets

AI / Machine Learning: Retrieval-Augmented Generation (RAG), LangChain, Vector Databases (Pinecone), Hugging Face Transformers, Google Gemini API, NLP, OpenCV, MediaPipe, YOLOv8, Convolutional Neural Networks (CNN), Object Detection, Real-Time Inference

Frontend: Next.js (App Router), React, Tailwind CSS, Responsive Design, Server-Side Rendering (SSR)

Databases & Cloud: PostgreSQL, MySQL, Supabase, Database Design, Query Optimization

DevOps & Tools: Docker, GitHub Actions, CI/CD Pipelines, Linux, Git, Vercel, Agile Development, Unit Testing, ROS 2, Gazebo

Projects

Lumina — Cloud-Native API Gateway & Reverse Proxy

2026

Rust, Tokio, Axum, Hyper

github.com/Bhargi777/Lumina

- Engineered a high-performance, cloud-native API gateway and reverse proxy in Rust using asynchronous, non-blocking I/O with Tokio for concurrent request handling at scale.
- Designed modular middleware and routing layers (load distribution, request forwarding, service abstraction) enabling pluggable backend integration and horizontal scalability.
- Applied memory-safe systems programming, zero-cost abstractions, and async runtime tuning to minimize latency in the request path.

AI Document RAG Chat — Conversational Document Intelligence Platform

2026

Next.js, FastAPI, LangChain, Pinecone, Docker

[GitHub](#)

- Built a production-ready retrieval-augmented generation (RAG) platform enabling semantic search and conversational Q&A over PDF documents.
- Architected a monorepo integrating FastAPI microservices, LangChain orchestration, Pinecone vector embeddings, and a Next.js frontend, fully containerized with Docker.
- Implemented document ingestion, text chunking, embedding generation, and context-aware LLM prompting for accurate grounded responses.

Real-Time Behavioral Intelligence System (RBIS)

2026

Python, OpenCV, MediaPipe, Computer Vision

[GitHub](#)

- Developed a real-time, multi-person computer vision pipeline detecting posture, gestures, attention state, and interaction events from live video streams.
- Implemented persistent multi-object tracking and temporal behavioral modeling, surfacing insights through an analytics dashboard.
- Optimized frame-level inference throughput for low-latency processing on commodity hardware.

Vehicle Headlamp Glare Detection
Python, YOLOv8-Nano, ROS 2, Gazebo

2026
GitHub

- Built an end-to-end object detection pipeline: trained a YOLOv8-Nano headlamp detection model on Apple Silicon and deployed it as a ROS 2 inference node validated in Gazebo simulation.
- Bridged deep learning and robotics tooling for an ADAS-style automotive perception use case.

RepoForge — AI Developer Documentation Tool
TypeScript, Next.js, LLM APIs

2026
GitHub

- Built an AI-powered developer tool that automatically generates professional README files, technical documentation, and architecture diagrams from any GitHub repository.
- Designed repository parsing and prompt-engineering workflows to produce accurate, structured documentation output.

INIT Logistics — QR-Based Attendance Platform
Next.js, TypeScript, Supabase

2025
GitHub

- Developed a full-stack QR code attendance tracking system with real-time live scanning and cloud synchronization, deployed for club events with 100+ attendees.
- Implemented real-time database sync with Supabase (PostgreSQL) and a responsive scanning UI for on-ground event operations.

Open Source & Engineering Practice

- Maintain 30+ public repositories spanning AI infrastructure, backend systems, computer vision, robotics, and developer productivity tooling.
- Practice production engineering workflows: version control (Git), pull-request reviews, GitHub Actions CI/CD, automated testing, and technical documentation.
- Implemented core data structures and algorithms in C++ (segment trees, graph algorithms, dynamic programming, minimax with memoization) applied in projects such as a segment-tree-backed restaurant search engine and a Java Swing AI game.

Leadership & Extracurriculars

INIT Club, Amrita Vishwa Vidyapeetham
Head of Logistics & Finance

2025 – Present
Coimbatore, India

- Led budgeting, logistics, and operational planning for technical workshops and events with 100+ attendees.
- Built and deployed internal tooling (QR attendance, event management systems) used in live club operations; co-organized a Git workshop teaching version control to beginners.

National Service Scheme (NSS)
Student Coordinator

2024 – Present
Amrita Vishwa Vidyapeetham

- Coordinated volunteers and institutional outreach programs; automated attendance tracking and certificate generation/distribution with self-built web utilities.

Certifications

- IBM SkillsBuild — Getting Started with Artificial Intelligence (2026)